B.Tech Degree VIII Semester Examination April 2011

ME 802 MACHINE DESIGN II

(2002 Scheme)

Time: 3 Hours | Maximum Marks: 100

(Design Data Book is permitted)
(Assume reasonable data wherever necessary)

I. (a) Explain *any two* types of friction clutches. (12)
(b) Derive an expression for braking torque in an internal expanding brake. (13)

OR

II. (a) Explain the chordal action in chains. (5)
(b) Two shafts whose centres are 100 cm apart are connected by a V – belt drive. The driving pulley is supplied with 125 HP and has an effective dia. of 30 cm. It runs at 1000 rpm while the driven pulley runs at 375 rpm. The angle of groove on the pulley is 40°. Permissible tension in 4 square cm. cross sectional area belt is 3 N/cm^2. The material of the belt weighs 1.11 gm/cm^3. The driven pulley is overhung, the distance of the centre from the nearest bearing being 20 cm. The coefficient of friction between belt and pulley is 0.28. Find the number of belts required and diameter of driven pulley shaft if permissible shear stress is 50 N/cm^2 (20)

III. (a) Describe various systems of gear teeth. (10)
(b) Explain design procedure for spur gears. (10)
(c) Explain interference in involute profiles. (5)

OR

IV. (a) Explain different types of bevel gears. (10)
(b) A pair of helical gears with 30° helix angle is used to transmit 15 KW at 10,000 rpm from the pinion shaft with a velocity ratio of 5 :1. The gear is made of phosphor bronze having static strength 75 N/mm^2 and the pinion is made of hardened steel of static strength 120 N/mm^2. Find the module, pitch dia, and face width for 20° full depth involute teeth. (15)

V. (a) Design a journal bearing for a 20 MW, 1800 rpm steam turbine which is supported by two bearings. Consider an operating temperature of 58° C. (15)
(b) Describe hydrodynamic lubricated bearing. (10)

OR

VI. (a) Explain rolling contact bearings. (9)
(b) Describe bearing characteristic number and bearing modulus. (8)
(c) Explain the properties to be considered in selection of bearing material. (8)

VII. (a) Explain design considerations for forgings. (10)
(b) Explain design recommendations for turned parts. (7)
(c) Describe design considerations for machined round holes. (8)

OR

VIII. (a) Explain design recommendations for castings. (10)
(b) Describe design considerations for welded parts. (5)
(c) Explain the procedure for preparation of working drawings. (10)

***